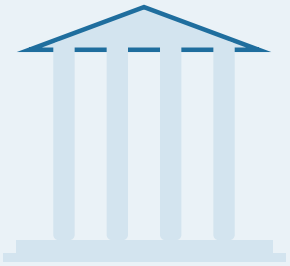


RETAIL BANK TRANSACTION ANALYSIS

POWER BI PROJECT REPORT

Turning banking data into clear insights through interactive dashboards.

Project Type	Data Analytics & Business Intelligence
Primary Tool	Microsoft Power BI
Datasets	8 banking datasets
Dashboard Pages	5 interactive pages
Prepared By	Mahesh Nasarla

1. Abstract and Introduction

This project analyzes retail banking data using Microsoft Power BI. The data covers customers, accounts, branches, transactions, loans, loan payments, and cards. The aim is to understand banking activity and present useful findings through interactive dashboards.

Banks work with information from many different areas. When this information is stored in separate datasets, it can be difficult to understand the complete picture. This project brings the datasets together in one Power BI report so users can explore banking operations more easily.

Project purpose

The report provides a visual view of customer behaviour, account activity, transaction patterns, loan repayment performance, and card usage. It is designed to make important information easier to explore and understand.

Expected value

The dashboards help users identify patterns, notice areas that may need attention, and use data to support business discussions. The report is an analytical aid and does not make banking decisions automatically.

2. Objectives, Scope and Tools

Project objectives

• Analyze customer details and activity.

• Understand account usage and branch performance.

• Identify transaction patterns and trends.

• Monitor loan amounts, outstanding balances, and delayed payments.

• Analyze card usage and customer engagement.

• Present findings through interactive dashboards.

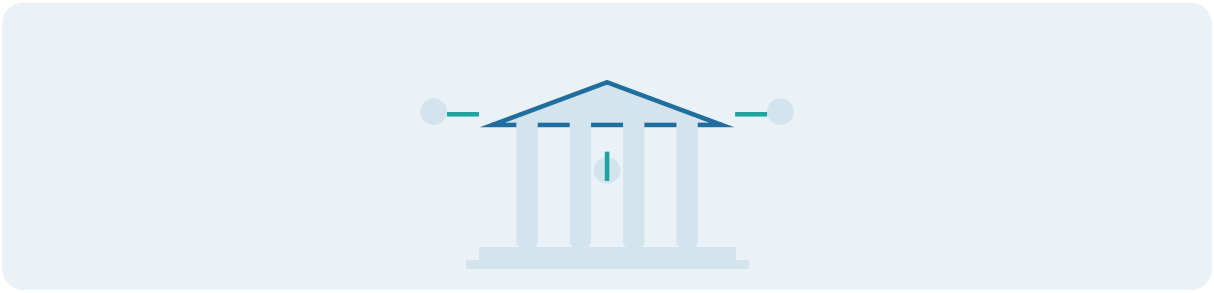
Project scope

The analysis covers eight banking datasets and five dashboard pages. The work includes data preparation, data modelling, DAX calculations, visualization, and business interpretation.

Tools and technologies

Tool	How it was used
Microsoft Power BI	Build the report and interactive dashboards.
Power Query	Clean and transform the source datasets.
DAX	Create measures, calculated columns, and KPIs.
Data modelling	Connect related tables for combined analysis.
Time intelligence	Support analysis across dates and time periods.

3. Dataset Description



The project uses eight datasets containing information from different areas of retail banking. The tables are connected in Power BI where suitable relationships exist.

Dataset	Size	What it contains
Customers	500	Customer details and profile information.
Accounts	700	Account information and balances.
Branches	40	Bank branch information.
Transactions	5,000	Customer transaction records.
Loans	300	Loan details and loan information.
Loan Payments	2,000	Loan repayment records.
Cards	600	Customer card information.
Date Table	Calendar	Date fields used for time-based analysis.

Dataset coverage

The datasets cover customer, account, branch, transaction, loan, repayment, and card information. The Date Table supports date-based reporting. The record counts above describe the supplied project datasets.

4. Data Preparation and Modelling

Data cleaning and transformation

Power Query was used to prepare the datasets before analysis. The main activities included:

- Checking missing values and reviewing how they should be handled.
- Correcting column data types.
- Standardizing column names and categories.
- Reviewing data consistency.
- Preparing the transformed tables for the Power BI model.

Data modelling

The cleaned tables were loaded into Power BI and relationships were created between relevant tables. This allows information from different banking areas to be analyzed together. A separate Date Table was included to support date filtering and time-based calculations.

DAX calculations

DAX measures and calculated columns were used to calculate business metrics and support the visuals. Examples include total customers, total accounts, active account percentage, total transaction amount, average transaction amount, total loan amount, outstanding loan balance, delayed payment records, and card activity metrics.

Why these steps matter

Data preparation improves consistency, data modelling connects related information, and DAX provides reusable calculations for the report. Together, these steps form the base of the Power BI dashboards.

5. Dashboard Pages: Customers and Accounts

Dashboard 1 — Customer Segmentation and Activity Analysis

This page focuses on the customer base. It helps explore customer segments, customer activity, and the distribution of customers across available categories. It can be used to understand the types of customers represented in the dataset and compare activity across groups.

Questions this page helps answer

- How many customers are included in the dataset?
- How are customers distributed across segments?
- What does the customer activity status look like?
- Are there differences between customer groups?

Dashboard 2 — Account Usage and Branch Performance

This page focuses on account balances, account activity, interest rates, and branch-level information. It helps compare account activity and understand how accounts are distributed across branches.

Questions this page helps answer

- How many accounts are active?
- What is the total account balance?
- How are accounts distributed across branches?
- What is the average interest rate?

6. Dashboard Pages: Transactions and Loans

Dashboard 3 — Transaction Trends and Pattern Analysis

This page analyzes transaction amounts, transaction types, customer activity, and trends over time. It helps users explore how transaction activity changes and which transaction categories contribute to the overall amount.

Questions this page helps answer

- What is the total transaction amount?
- What is the average and highest transaction amount?
- How do transaction values change over time?
- How many customers are actively making transactions?

Dashboard 4 — Loan Performance and Repayment Analysis

This page focuses on the loan portfolio and repayment records. It displays loan amounts, outstanding balances, payment activity, and delayed payments to help monitor repayment behaviour.

Questions this page helps answer

- What is the total loan amount?
- How much loan balance is outstanding?
- How many payment records are delayed?
- How does loan performance differ by loan type?

7. Dashboard Page: Cards and Key Findings

Dashboard 5 — Card Usage and Product Engagement

This page analyzes customer card activity, card types, networks, credit limits, outstanding balances, and rewards where available. It helps understand card engagement and identify areas where customer usage could be improved.

Questions this page helps answer

- How many cards are active or inactive?
- Which card types or networks are used?
- What are the credit limits and outstanding balances?
- How can card activation and engagement be improved?

Key findings from the current analysis

Observation	Result	Meaning
Active accounts	497 of 700 (about 71%)	Most accounts in the dataset are marked active.
Delayed loan payment records	500 of 2,000 (25%)	One quarter of payment records are marked as delayed.
Loan portfolio	About ■72.21 million	Represents the total loan amount identified in the analysis.
Outstanding loan balance	About ■53.44 million	Shows the remaining loan balance reported in the analysis.

Note: The 25% figure refers to delayed loan payment records, not to the number of loans. Values should be checked against the final PBIX report before formal submission.

8. Recommendations and Learning Outcomes

Business recommendations

1. Improve card activation: Use customer awareness campaigns and suitable offers to encourage card activation.
2. Monitor delayed payments: Review delayed repayment records and follow up with customers who may need support.
3. Improve customer engagement: Use customer segments to understand needs and provide relevant banking services.
4. Review account activity: Monitor inactive accounts and explore possible reasons for low usage.
5. Track transaction trends: Review transaction patterns regularly to understand changes in customer activity.

Learning outcomes

This project provided practical experience in Power Query data cleaning, data modelling, creating relationships, writing DAX measures and calculated columns, building KPI cards, using slicers and filters, applying time intelligence, and designing interactive dashboards.

Challenges addressed

The main tasks included working with multiple datasets, checking missing values and data types, creating suitable table relationships, writing calculations, choosing appropriate visuals, and presenting findings in a clear way.

9. Conclusion

The Retail Bank Transaction Analysis project demonstrates how Microsoft Power BI can be used to turn banking data into meaningful information. The project brought together eight datasets and used Power Query, data modelling, DAX, and interactive visualizations to analyze different areas of retail banking.

The five dashboard pages provide views of customer activity, account usage, transaction patterns, loan performance, and card engagement. The findings and recommendations show how dashboards can help users identify trends and areas that may need further attention.

Overall, this project strengthened practical skills in the Power BI reporting process, from preparing datasets to building dashboards and communicating business insights.

PROJECT SUMMARY

Datasets	Dashboard Pages	Main Tool
8	5	Power BI

Prepared by: Mahesh Nasarla

Project: Retail Bank Transaction Analysis | Microsoft Power BI